

Gen AI & Agentic AI Interview Questions

Beginner

1. What is Generative AI, explain how it differs from traditional AI.
2. What are AI agents and what is Agentic AI?
3. How do Generative AI, AI Agents, and Agentic AI differ? What is their core purpose?
4. What are the main steps to follow in order to build an Agentic AI application?
5. What are Large Language Models (LLMs), and what is their basic architecture?
6. What are some real-world applications of Generative AI and Agentic AI?
7. Explain the concept of prompt engineering and describe zero-shot, one-shot, and few-shot prompting.
8. What are common challenges of using Generative AI?
9. Describe Retrieval Augmented Generation (RAG) architecture, its purpose, and its key benefits.
10. What are some initial ethical concerns associated with Generative AI and Agentic AI?

Intermediate

1. Explain the transformer architecture and elaborate the attention mechanism.
2. What is context window, temperature, top-p, and top-k in LLMs?
3. What is a Vector Database and why use it with LLMs?
4. What is Euclidean distance and Cosine similarity in vector databases?
5. What is the purpose of the LangChain framework, and how do its core components like Prompt Templates, Chains, and Output Parsers facilitate LLM application development?
6. When might you use a Multi-agent system?
7. Discuss the Model Context Protocol (MCP) and the Agent2Agent (A2A) Protocol, explaining their purposes, key capabilities, and how they relate in the evolving AI agent ecosystem.
8. Explain the purpose of LLM fine-tuning, and when should we opt for it instead of RAG?
9. What is Low-Rank Adaptation (LoRA) in LLM fine-tuning?
10. What is quantization and what is QLoRA?

Advanced

1. How can we evaluate the performance of Agentic AI systems?
2. You have a legal document that's 100 pages long. Explain strategies to get accurate Q&A from an LLM with a 4K-token context window.
3. Explain how you'd design metadata filters in ChromaDB (or other similar vectorDB) to let users search a legal corpus by jurisdiction and case type.
4. You need to generate product descriptions for 1,000 SKUs. Walk me through how you'd design your prompts to ensure consistent tone and accuracy.
5. When building a Retrieval-Augmented Generation system, what are the pros and cons of fetching long passages vs. short passages from your knowledge base?
6. Imagine you're building a customer-support chatbot. How would you decide whether to use a traditional rule-based system or a generative model like an LLM? What trade-offs would you consider?
7. With generative and agentic AI evolving rapidly, how would you architect your platform to accommodate new LLM variants, protocols, and regulatory changes over the next 2–3 years?
8. Explain how 4-bit quantization can impact both inference latency and model accuracy. In which scenarios might you avoid aggressive quantization?
9. Your company has a proprietary dataset of 10,000 support transcripts. Would you choose LoRA, QLoRA, or full-model fine-tuning? Justify your decision in terms of cost, performance, and maintainability.
10. Your enterprise uses Google Drive, Jira, and Confluence. Sketch how you'd build an MCP server to let any LLM-based assistant query across those systems.